

TrES-2b

R-0087

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_181018 · created 2026-07-15T21:46:26+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458409.68069 +/- 0.00061 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1389 +/- 0.0085
Transit depth (Rp/Rs)^2	1.93 +/- 0.24 [%]
Orbital Inclination (inc)	84.29 +/- 0.29
Airmass coefficient 1 (a1)	1.1986 +/- 0.0095
Airmass coefficient 2 (a2)	-0.1089 +/- 0.0039
Scatter in the residuals of the lightcurve fit is	0.78 %
Best Comparison Star	None
Optimal Aperture	4.95
Optimal Annulus	12.88
Transit Duration (day)	0.0826 +/- 0.004

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1389 ± 0.0085 vs 0.1254 (deviation 1.59σ)
Duration fit vs published	0.0826 d vs 0.0746 d (deviation 2.0σ)
Mid-time O–C	-0.2 min
Depth SNR	16.3
Residual scatter	0.78 %

Light curve

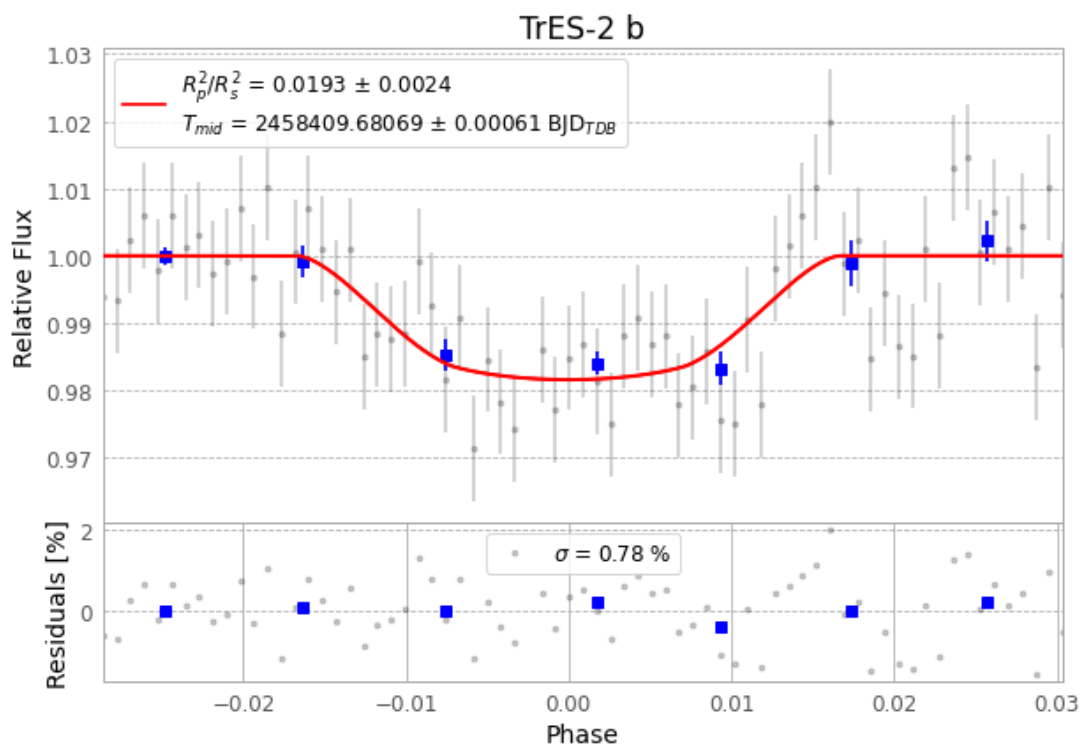


Figure 1 — FinalLightCurve_TrES-2 b_2018-10-17.png (SHA-256 37c605e1f211a17b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [293, 358], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ec1c0128d80d59a1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

71 FITS frames (47 MB), TRES-2181018023612.FITS → TRES-2181018060612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-181018.log (b86f191615ed...), FinalLightCurve_TrES-2 b_2018-10-17.png (37c605e1f211...), FinalLightCurve_TrES-2 b_2018-10-17.csv (8c104d8e7561...)

Compute resources

Wall time 125.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 21:46Z.